C

ORDINARY DIFFERENTIAL EQUATIONS • UNIT 9.01

Numerical Methods: Euler, RK2, RK4 & Stability

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

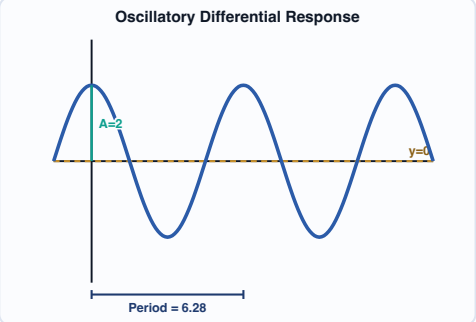
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 D	Q3 D	Q4 A	Q5 A
Q6 A	Q7 D	Q8 A	Q9 D	Q10 D
Q11 A	Q12 B	Q13 A	Q14 B	Q15 A
Q16 D	Q17 D	Q18 A	Q19 D	Q20 B
Q21 B	Q22 A	Q23 D	Q24 B	Q25 D

Numerical Methods: Euler, RK2, RK4 & Stability Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



- **A** 39 ✓
- **B** 24
- **C** 42
- **D** 53

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 39\lambda + 350 = 0$, the sum of roots is 39.

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

B $e^2 - 1$

C $3 - 2e^{-2}$

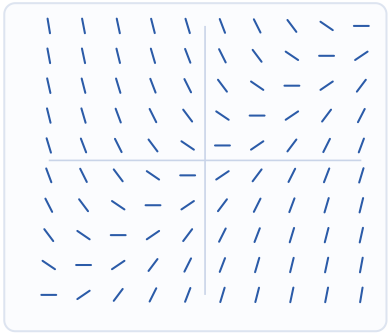
D $3 - 2e^{-1}$ ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is a linear first-order ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 2$ and $Q(x) = 6$. The integrating factor $\mu(x)$ is given by $\mu(x) = e^{\int P(x)dx}$.
Calculating the integrating factor: $\mu(x) = e^{\int 2dx} = e^{2x}$.
Multiplying the differential equation by the integrating factor: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$.
The left side is the derivative of the product $y\mu(x)$: $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$.
Integrating both sides with respect to x: $ye^{2x} = \int 6e^{2x}dx$.
$ye^{2x} = 6 \cdot \frac{1}{2}e^{2x} + C$.
$ye^{2x} = 3e^{2x} + C$.
Solving for y: $y = 3 + Ce^{-2x}$.
Using the initial condition $y(0) = 1$ to find C: $1 = 3 + Ce^{-2(0)}$.
$1 = 3 + C$.
$C = 1 - 3 = -2$.
So, the particular solution is $y(x) = 3 - 2e^{-2x}$.
Now, we need to find $y(1)$: $y(1) = 3 - 2e^{-2(1)} = 3 - 2e^{-2}$.
This matches choice C.

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D **1.220** ✓

WORKED SOLUTION & VERIFICATION

Using Euler's method $y_{n+1} = y_n + h(x_n + y_n)$ with $x_0 = 0, y_0 = 1, h = 0.1$:
1. $y_1 = 1 + 0.1(0 + 1) = 1.100$ at $x_1 = 0.1$.
2. $y_2 = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.12 = 1.220$ at $x_2 = 0.2$.

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

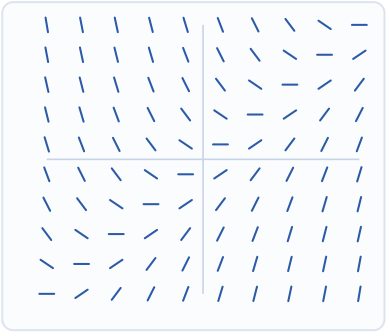
- **A** 5 ✓
- **B** -5
- **C** 6
- **D** -6

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' - 5y' + 6y = 0$. The auxiliary equation for a second-order linear homogeneous differential equation of the form $ay'' + by' + cy = 0$ is $ar^2 + br + c = 0$. In this case, $a = 1$, $b = -5$, and $c = 6$. So the auxiliary equation is $r^2 - 5r + 6 = 0$. Let the roots of this quadratic equation be r_1 and r_2. According to Vieta's formulas, for a quadratic equation $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2$ is equal to $-\frac{b}{a}$. In our auxiliary equation $r^2 - 5r + 6 = 0$, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $r_1 + r_2 = -\frac{-5}{1} = 5$.

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



- **A** 7 ✓
- **B** 6
- **C** 8
- **D** 5

WORKED SOLUTION & VERIFICATION

The differential equation is $\frac{dy}{dx} = 2x$. This is a separable equation. We can rewrite it as $dy = 2x\,dx$. Integrating both sides gives $\int dy = \int 2x\,dx$, which results in $y = x^2 + C$, where C is the constant of integration. We are given the initial condition $y(0) = 3$. Substituting these values into the general solution, we get $3 = (0)^2 + C$, so $C = 3$. Therefore, the particular solution is $y = x^2 + 3$. We need to find the value of $y(2)$. Substituting $x = 2$ into the particular solution, we have $y(2) = (2)^2 + 3 = 4 + 3 = 7$.

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

- **A** $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$ ✓
- **B** $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$
- **C** $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$
- **D** $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

WORKED SOLUTION & VERIFICATION

We are given the differential equation $y'' - xy = 0$ and assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$. \newline Differentiating twice with respect to x, we get: $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$. \newline Substituting these into the differential equation: $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$. \newline We can rewrite the second term as $\sum_{n=0}^{\infty} c_n x^{n+1}$. To combine the series, we need the powers of x to be the same. Let $k = n - 2$ in the first series, so $n = k + 2$. When $n = 2, k = 0$. The first series becomes $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$. \newline Let $k = n + 1$ in the second series, so $n = k - 1$. When $n = 0, k = 1$. The second series becomes $\sum_{k=1}^{\infty} c_{k-1} x^k$. \newline So the equation is $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline We can separate the $k = 0$ term from the first sum: $2(1)c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline For this equation to hold for all x, the coefficients of each power of x must be zero. \newline For $k = 0$: $2c_2 = 0$ implies $c_2 = 0$. \newline For $k \geq 1$: $(k+2)(k+1)c_{k+2} - c_{k-1} = 0$. \newline Thus, the recurrence relation is $c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$ for $k \geq 1$. Replacing k with n, we get $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$.

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

- **A** 44
- **B** 43
- **C** 24
- **D** 40 ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 40\lambda + 144 = 0$, the sum of roots is 40.

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

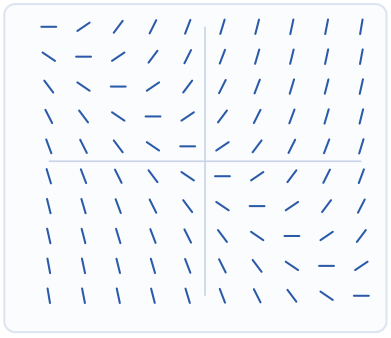
- **A** 170 ✓
- **B** 151
- **C** 173
- **D** 187

WORKED SOLUTION & VERIFICATION

Substituting $s = 20$ gives $\frac{170}{20-19} = 170$.

9 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. You are asked to approximate $y(0.2)$ using Euler's method. If the step size $h = 0.1$ is used, and it is known that the exact solution at $x = 0.2$ is $y(0.2) = 1.24$, determine the value of k such that $y(0.2)_{\text{Euler}} = 1.24 - k$.



A 0.03

B 0.04

C 0.01

D **0.02** ✓

WORKED SOLUTION & VERIFICATION

Euler's method for approximating the solution to $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$ and step size h is given by $y_{n+1} = y_n + hf(x_n, y_n)$.

In this problem, $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.1$.

We want to approximate $y(0.2)$ with $h = 0.1$. This requires two steps:

Step 1: From $x_0 = 0$ to $x_1 = 0.1$.

$y_1 = y_0 + hf(x_0, y_0)$

$y_1 = 1 + 0.1(0 + 1)$

$y_1 = 1 + 0.1(1)$

$y_1 = 1 + 0.1$

$y_1 = 1.1$

Step 2: From $x_1 = 0.1$ to $x_2 = 0.2$.

$y_2 = y_1 + hf(x_1, y_1)$

$y_2 = 1.1 + 0.1(0.1 + 1.1)$

$y_2 = 1.1 + 0.1(1.2)$

$y_2 = 1.1 + 0.12$

$y_2 = 1.22$

So, the approximation using Euler's method is $y(0.2)_{\text{Euler}} = 1.22$.

We are given that the exact solution is $y(0.2) = 1.24$ and $y(0.2)_{\text{Euler}} = 1.24 - k$.

Substituting the value of $y(0.2)_{\text{Euler}}$, we get:

$1.22 = 1.24 - k$

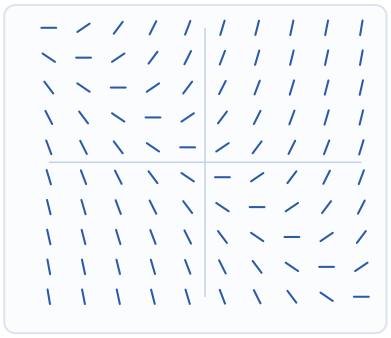
$k = 1.24 - 1.22$

$k = 0.02$

Therefore, the value of k is 0.02.

10 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size $h = 0.1$ to approximate the value of $y(0.2)$.



A 1.23

B 1.24

C 1.20

D **1.21** ✓

WORKED SOLUTION & VERIFICATION

The Euler method approximates the solution to an initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$ using the iterative formula $y_{n+1} = y_n + hf(x_n, y_n)$, where h is the step size and $x_{n+1} = x_n + h$.

Given $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.1$.

For the first step (approximating y(0.1)):

$x_1 = x_0 + h = 0 + 0.1 = 0.1$

$y_1 = y_0 + hf(x_0, y_0) = 1 + 0.1(0 + 1) = 1 + 0.1(1) = 1.1.$

For the second step (approximating y(0.2)):

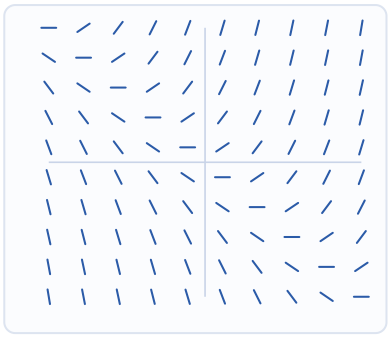
$x_2 = x_1 + h = 0.1 + 0.1 = 0.2$

$y_2 = y_1 + hf(x_1, y_1) = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.1(1.2) = 1.1 + 0.12 = 1.22.$

Therefore, the approximate value of $y(0.2)$ using the Euler method is 1.22.

11 MCQ

Consider the ordinary differential equation $\frac{dy}{dx} = x + y$, with the initial condition $y(0) = 1$. We will use Euler's method to approximate the value of $y(0.2)$ using a step size of $h = 0.1$. Let $y_0 = y(x_0)$, $y_1 = y(x_1)$, and $y_2 = y(x_2)$ be the approximations at $x_0 = 0$, $x_1 = 0.1$, and $x_2 = 0.2$ respectively.
Which of the following statements is TRUE regarding these approximations?



A $y_1 = 1.1$ **and the approximation for** $y(0.2)$ **is** 1.22 ✓

B $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.32

C $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.30

D $y_1 = 1.1$ and the approximation for $y(0.2)$ is 1.21

WORKED SOLUTION & VERIFICATION

Euler's method formula is given by $y_{n+1} = y_n + h \cdot f(x_n, y_n)$, where $f(x, y) = x + y$ and $h = 0.1$.
For the first step, from $x_0 = 0$ to $x_1 = 0.1$:
$y_1 = y_0 + h \cdot f(x_0, y_0)$
$y_1 = 1 + 0.1 \cdot (0 + 1)$
$y_1 = 1 + 0.1 \cdot 1$
$y_1 = 1 + 0.1 = 1.1$
So, the approximation for $y(0.1)$ is $y_1 = 1.1$.
For the second step, from $x_1 = 0.1$ to $x_2 = 0.2$:
$y_2 = y_1 + h \cdot f(x_1, y_1)$
$y_2 = 1.1 + 0.1 \cdot (0.1 + 1.1)$
$y_2 = 1.1 + 0.1 \cdot (1.2)$
$y_2 = 1.1 + 0.12 = 1.22$
Thus, the approximation for $y(0.2)$ is $y_2 = 1.22$. Therefore, the correct statement is that $y_1 = 1.1$ and the approximation for $y(0.2)$ is 1.22.

12 MCQ

The first-order differential equation $\frac{df}{dx} = 12x - 2f$ has its slope field plotted below. Determine the slope $\frac{df}{dx}$ at the point $A(6, 23)$.

A 38

B 26 ✓

C 3

D 29

WORKED SOLUTION & VERIFICATION

Evaluating at $(6, 23)$ gives $12(6) - 2(23) = 26$.

13 MCQ

The first-order differential equation $\frac{dp}{dt} = 22t - 5p$ has its slope field plotted below. Determine the slope $\frac{dp}{dt}$ at the point $P(7, 10)$.

- **A** 104 ✓
- **B** 94
- **C** 107
- **D** 126

WORKED SOLUTION & VERIFICATION

Evaluating at $(7, 10)$ gives $22(7) - 5(10) = 104$.

14 MCQ

For Euler's method, dividing the step size h by 10 reduces the global error by roughly what factor?

- **A** 100
- **B** 10 ✓
- **C** 10000
- **D** 20

WORKED SOLUTION & VERIFICATION

Global error $\propto h^1$, so the error shrinks by about $10^1 = 10$.

15 MCQ

What is the global truncation error of the improved Euler (Heun) method?

- **A** $O(h^2)$ ✓
- **B** $O(h)$
- **C** $O(h^3)$
- **D** $O(h^4)$

WORKED SOLUTION & VERIFICATION

The improved Euler (Heun) method is a method of order 2: its global error is $O(h^2)$.

16 MCQ

Use Euler's method with $h = \frac{1}{4}$ to approximate $y(\frac{1}{4})$ for $y' = -y$, $y(0) = 3$.

- **A** 3
- **B** $\frac{3}{2}$
- **C** $\frac{75}{32}$
- **D** $\frac{9}{4}$ ✓

WORKED SOLUTION & VERIFICATION

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(0, 3)$, gives $\frac{9}{4}$.

17 MCQ

For the improved Euler (Heun) method, dividing the step size h by 3 reduces the global error by roughly what factor?

A 3
B 27
C 81
D 9 ✓

WORKED SOLUTION & VERIFICATION

Global error $\propto h^2$, so the error shrinks by about $3^2 = 9$.

18 MCQ

Use Euler's method with $h = \frac{1}{4}$ to approximate $y(\frac{1}{4})$ for $y' = 3 - y$, $y(0) = 1$.

A $\frac{3}{2}$ ✓
B $\frac{23}{16}$
C 1
D 2

WORKED SOLUTION & VERIFICATION

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(0, 1)$, gives $\frac{3}{2}$.

19 MCQ

For the improved Euler (Heun) method, dividing the step size h by 10 reduces the global error by roughly what factor?

A 10
B 1000
C 10000
D 100 ✓

WORKED SOLUTION & VERIFICATION

Global error $\propto h^2$, so the error shrinks by about $10^2 = 100$.

20 MCQ

Use Euler's method with $h = \frac{1}{10}$ to approximate $y(\frac{1}{10})$ for $y' = 3 - y$, $y(0) = 1$.

A $\frac{119}{100}$
B $\frac{6}{5}$ ✓
C 1
D $\frac{7}{5}$

WORKED SOLUTION & VERIFICATION

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(0, 1)$, gives $\frac{6}{5}$.

21 MCQ

Use Euler's method with $h = \frac{1}{5}$ to approximate $y(\frac{6}{5})$ for $y' = 1 + y$, $y(1) = 3$.

A $\frac{97}{25}$

B $\frac{19}{5}$ ✓

C 3

D $\frac{23}{5}$

WORKED SOLUTION & VERIFICATION

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(1, 3)$, gives $\frac{19}{5}$.

22 MCQ

For Euler's method, dividing the step size h by 3 reduces the global error by roughly what factor?

A 3 ✓

B 6

C 9

D 81

WORKED SOLUTION & VERIFICATION

Global error $\propto h^1$, so the error shrinks by about $3^1 = 3$.

23 MCQ

Use Euler's method with $h = \frac{1}{5}$ to approximate $y(\frac{6}{5})$ for $y' = x$, $y(1) = 1$.

A 1

B $\frac{7}{5}$

C $\frac{61}{50}$

D $\frac{6}{5}$ ✓

WORKED SOLUTION & VERIFICATION

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(1, 1)$, gives $\frac{6}{5}$.

24 MCQ

Use Euler's method with $h = \frac{1}{2}$ to approximate $y(\frac{3}{2})$ for $y' = x^2$, $y(1) = 2$.

A $\frac{45}{16}$

B $\frac{5}{2}$ ✓

C 2

D 3

WORKED SOLUTION & VERIFICATION

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(1, 2)$, gives $\frac{5}{2}$.

25 MCQ

Use Euler's method with $h = \frac{1}{2}$ to approximate $y(\frac{1}{2})$ for $y' = -y$, $y(0) = 2$.

A 0

B $\frac{5}{4}$

C 2

D 1 ✓

WORKED SOLUTION & VERIFICATION

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(0, 2)$, gives 1.